

COE 1195: Advanced Digital Design

Lab 4 – 32-bits MIPS CPU

Milestone 3

Dr. Amr Mahmoud

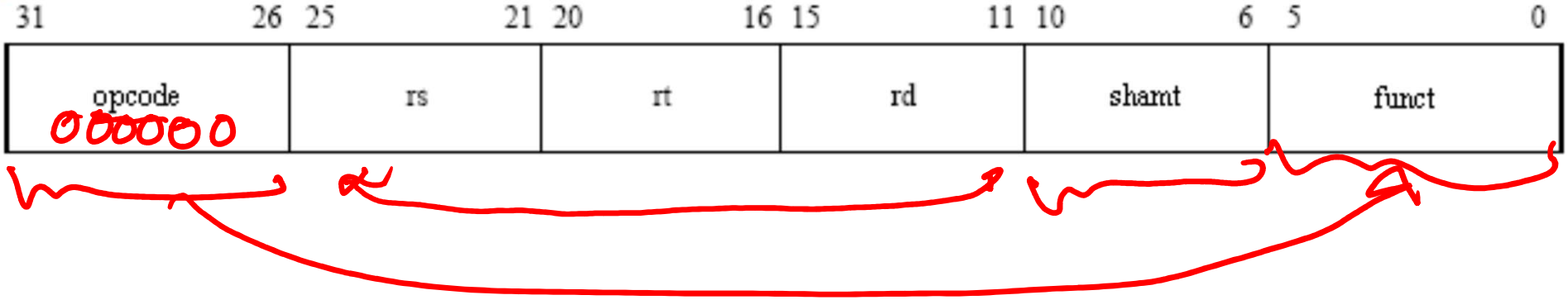
Milestone 3

1. Implement R-Type instructions

2. Implement J-Type Instructions

R-Type Instructions

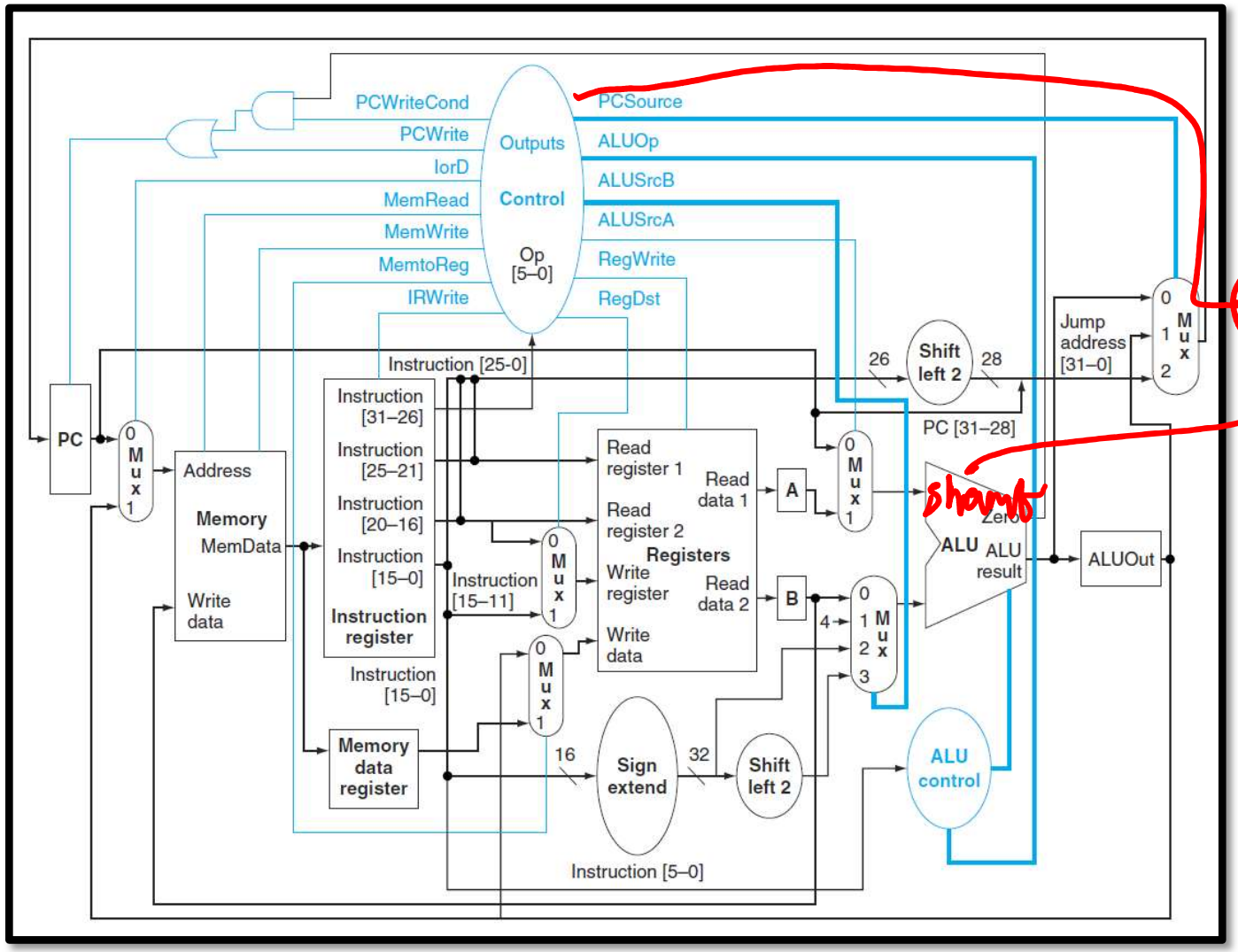
R-Type Instruction Format



R-Type Instructions

ADDI	ADDU	AND
ORI	SUB	SRA
SLLV	SLL	SLTI
LUI	SW	LW
LH	LB	CLO
BNE	BLTZAL	J
JR	MULTU	MFHI
MFLO		

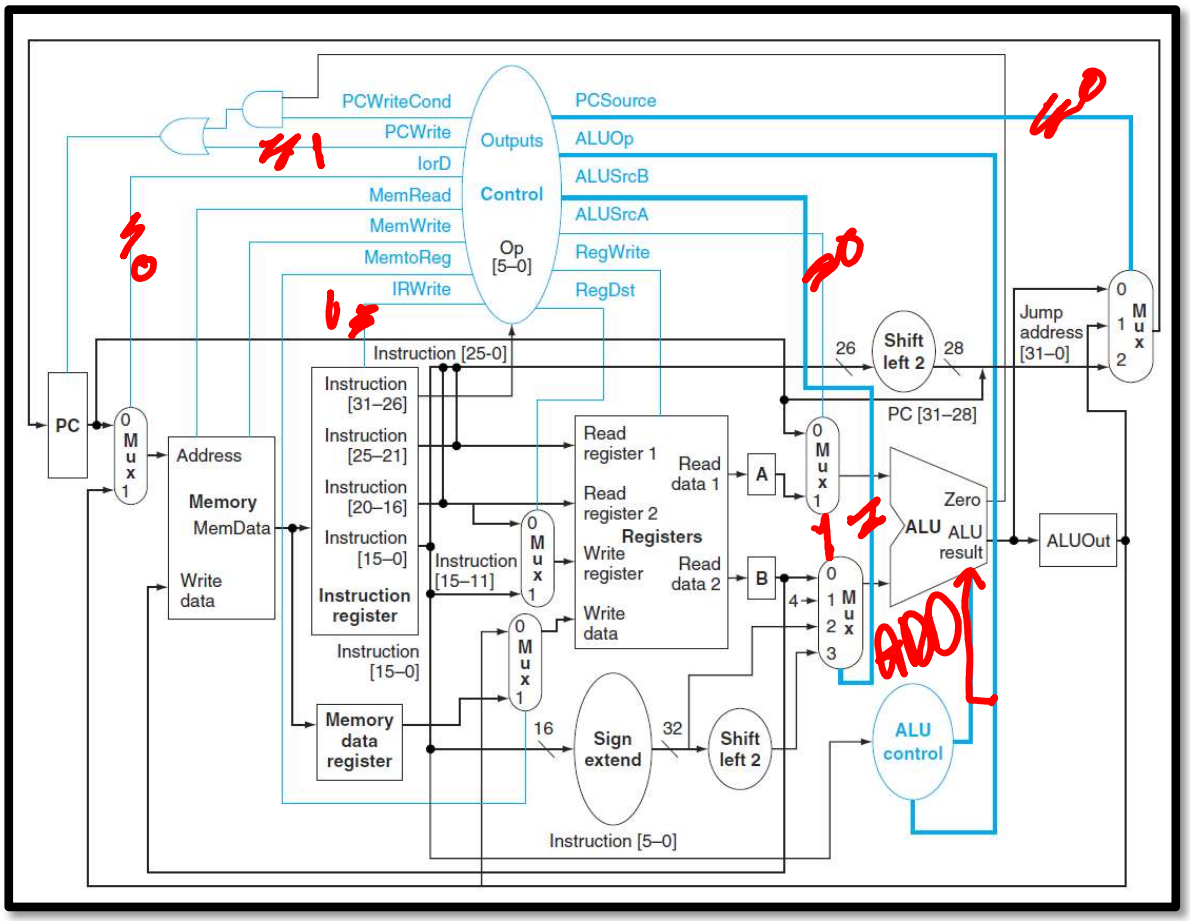
Milestone 3 – Implement R-Type instructions



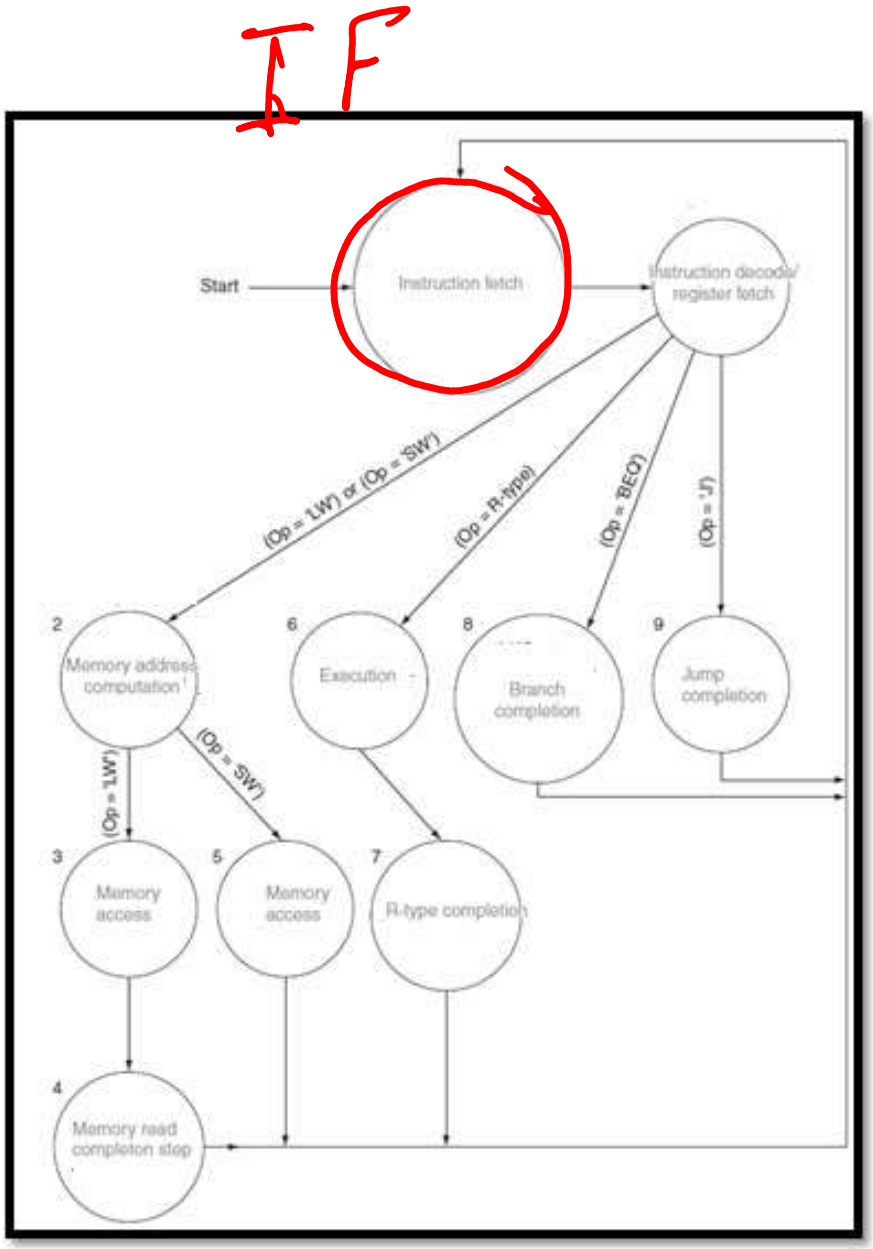
MIPS Multicycle CPU Datapath

Datapath and Control

MIPS Multicycle CPU FSM



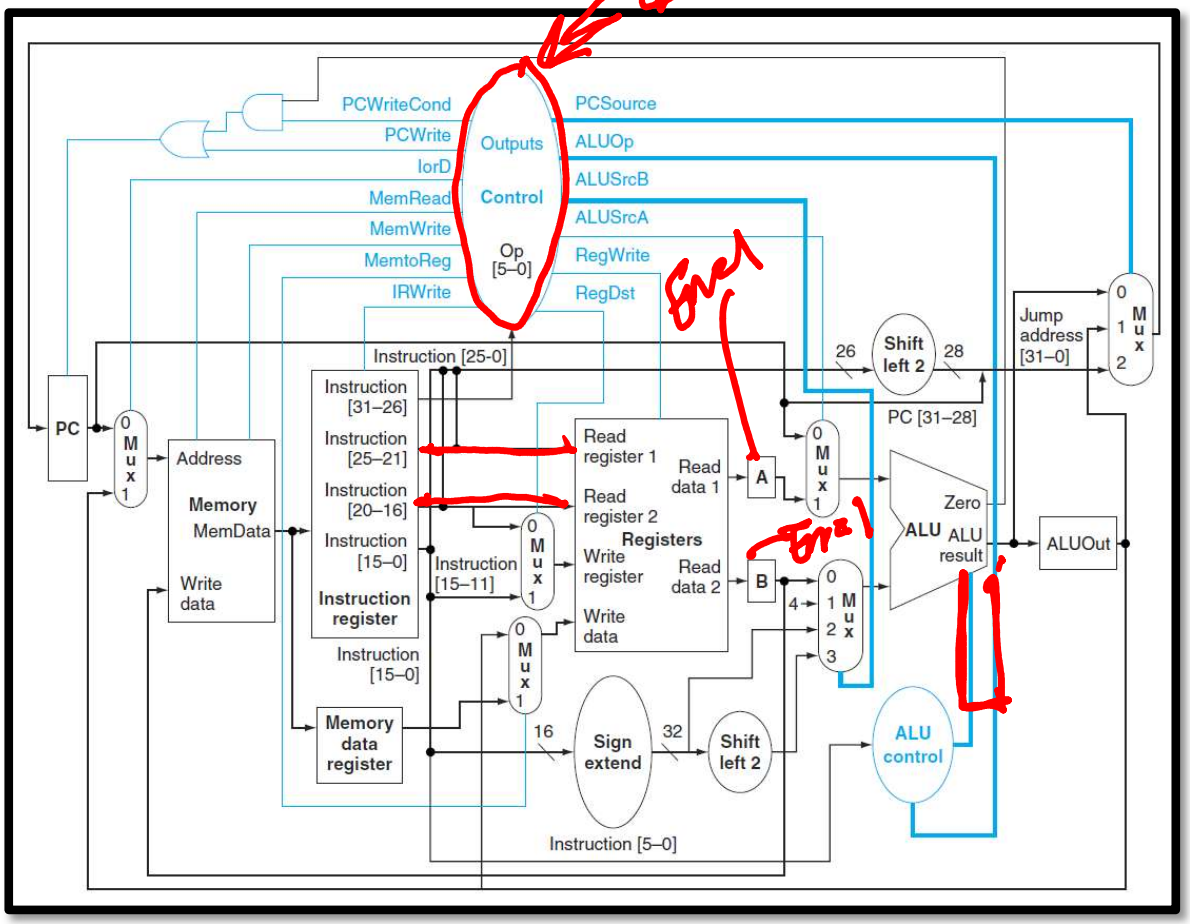
MIPS Multicycle CPU Datapath



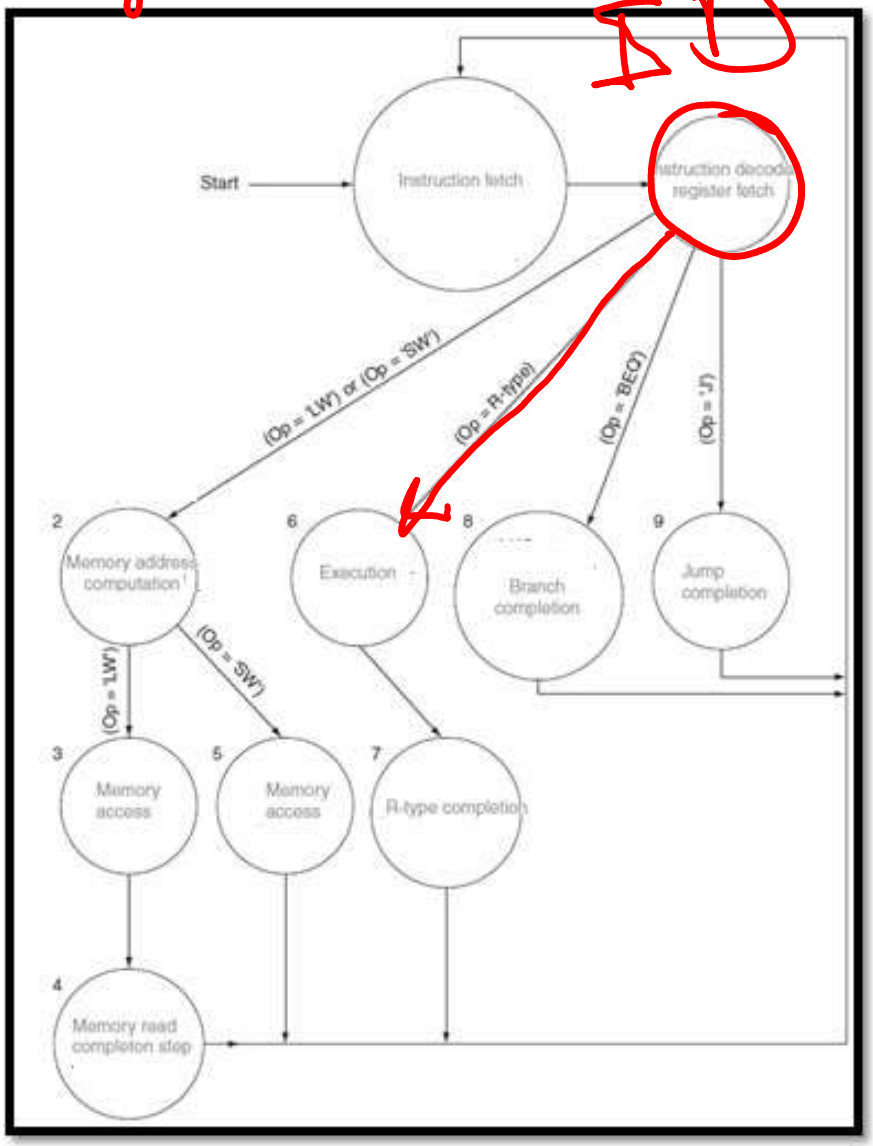
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



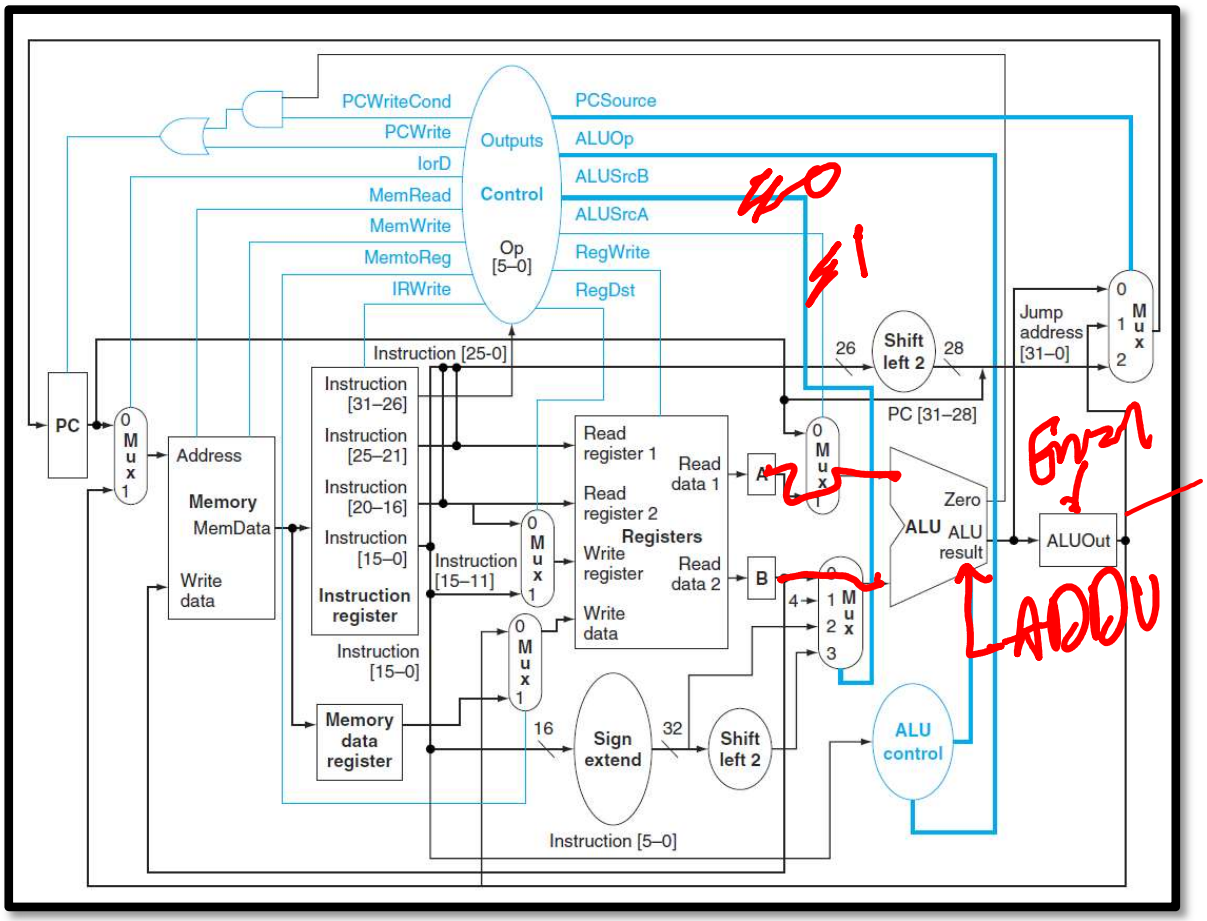
MIPS Multicycle CPU Datapath



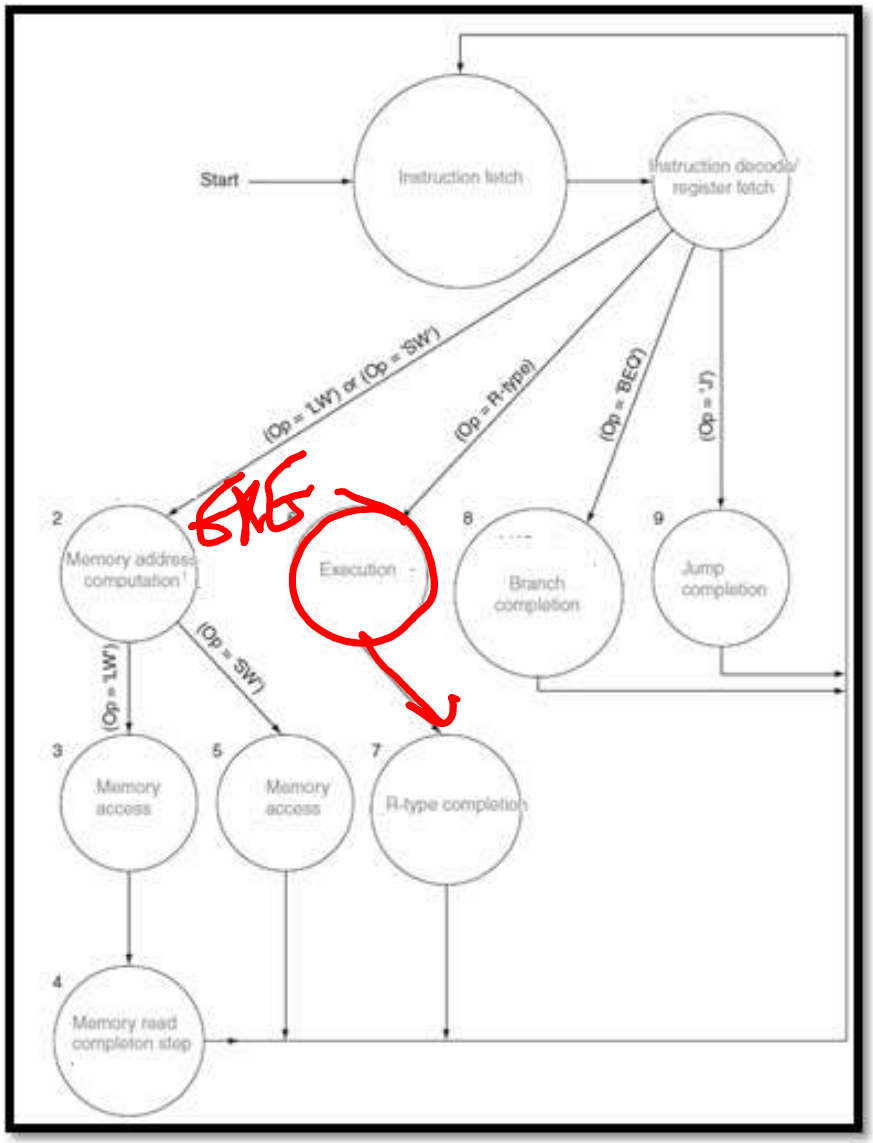
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



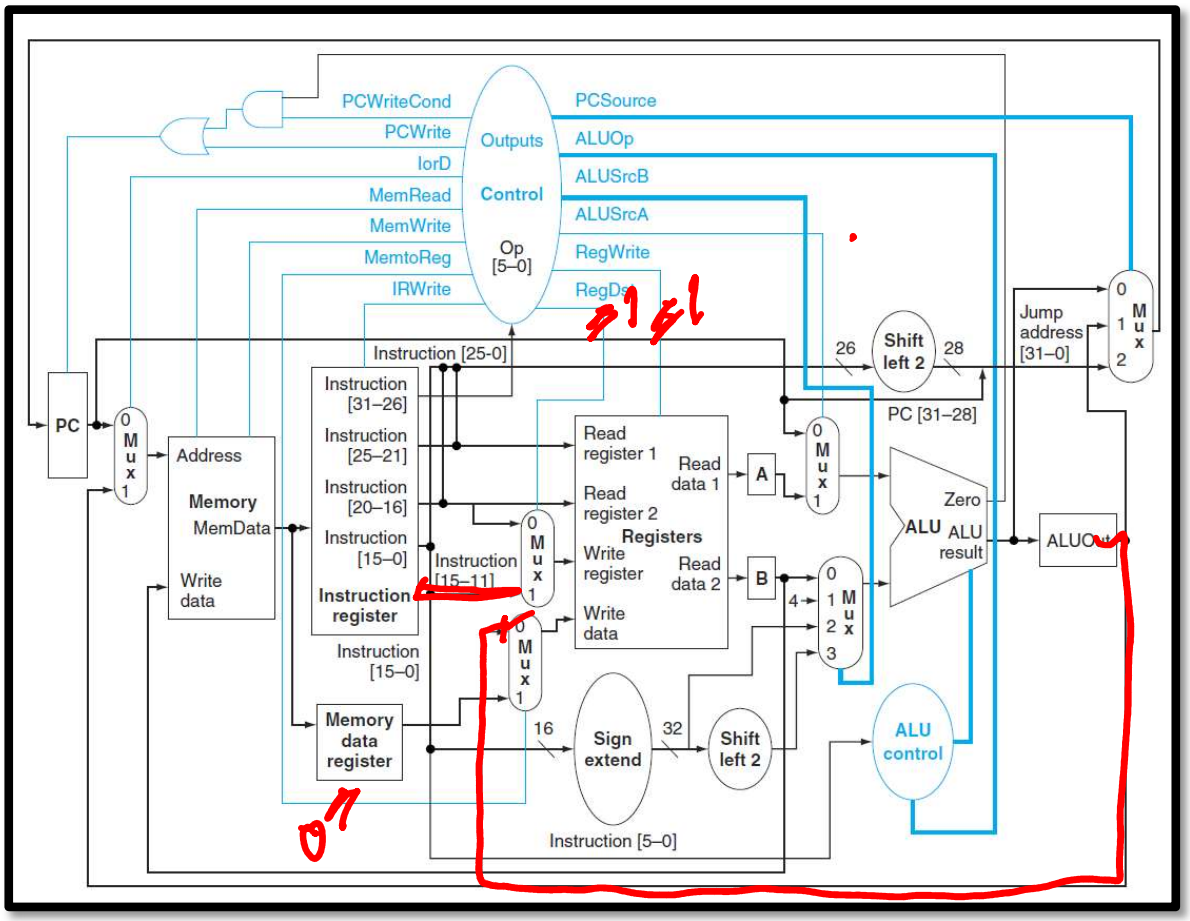
MIPS Multicycle CPU Datapath



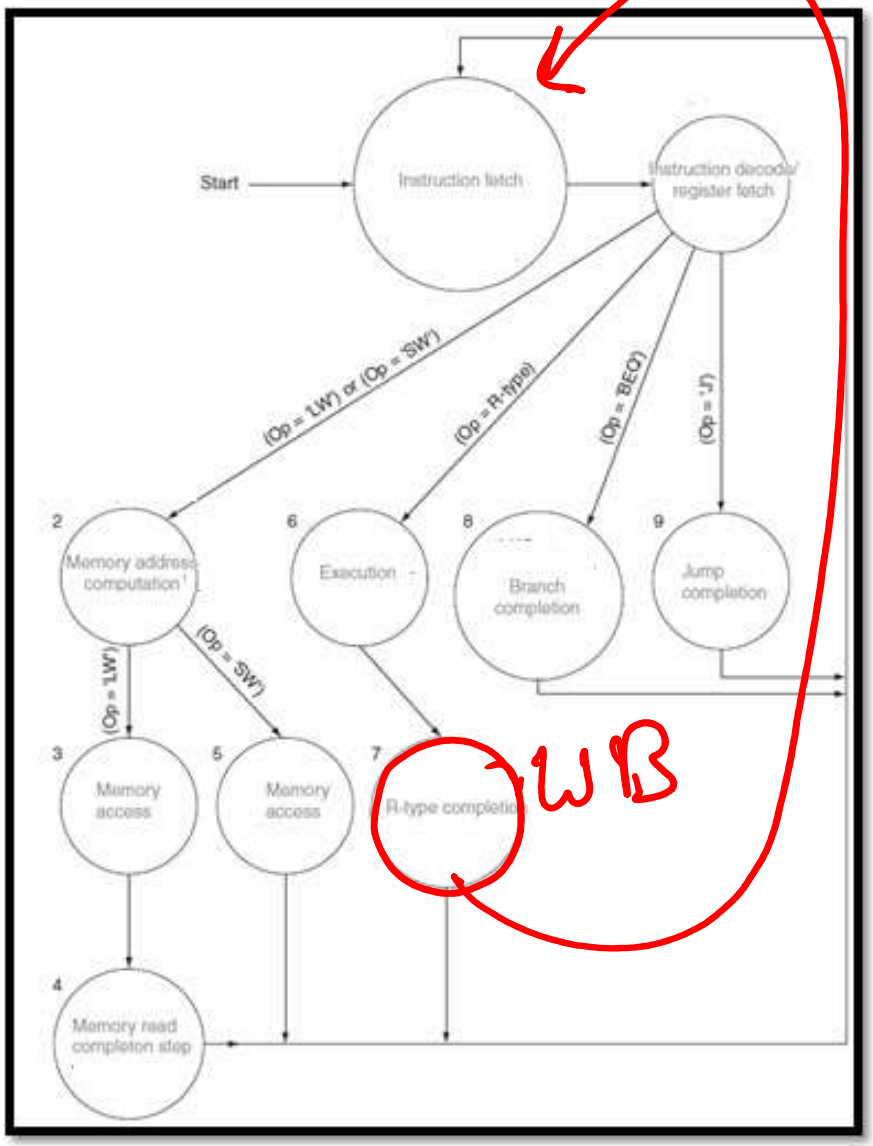
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



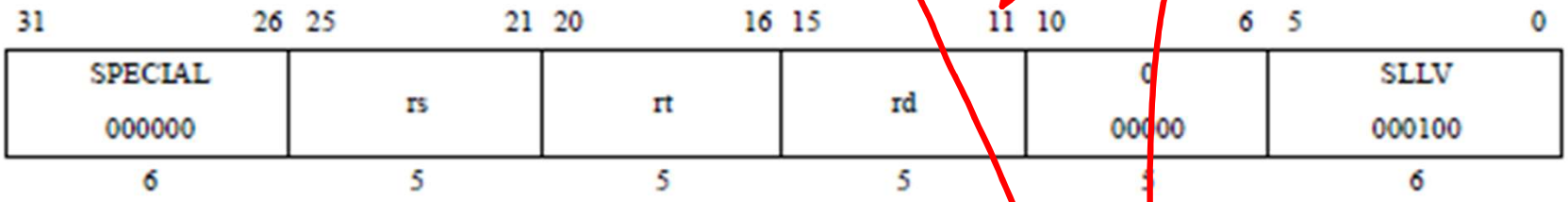
MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

SLLV

Shift Word Left Logical Variable — SLLV



Format: `SLLV rd, rt, rs` — MIPS32 (MIPS I)

Purpose: To left-shift a word by a variable number of bits

Description: `rd ← rt << rs`

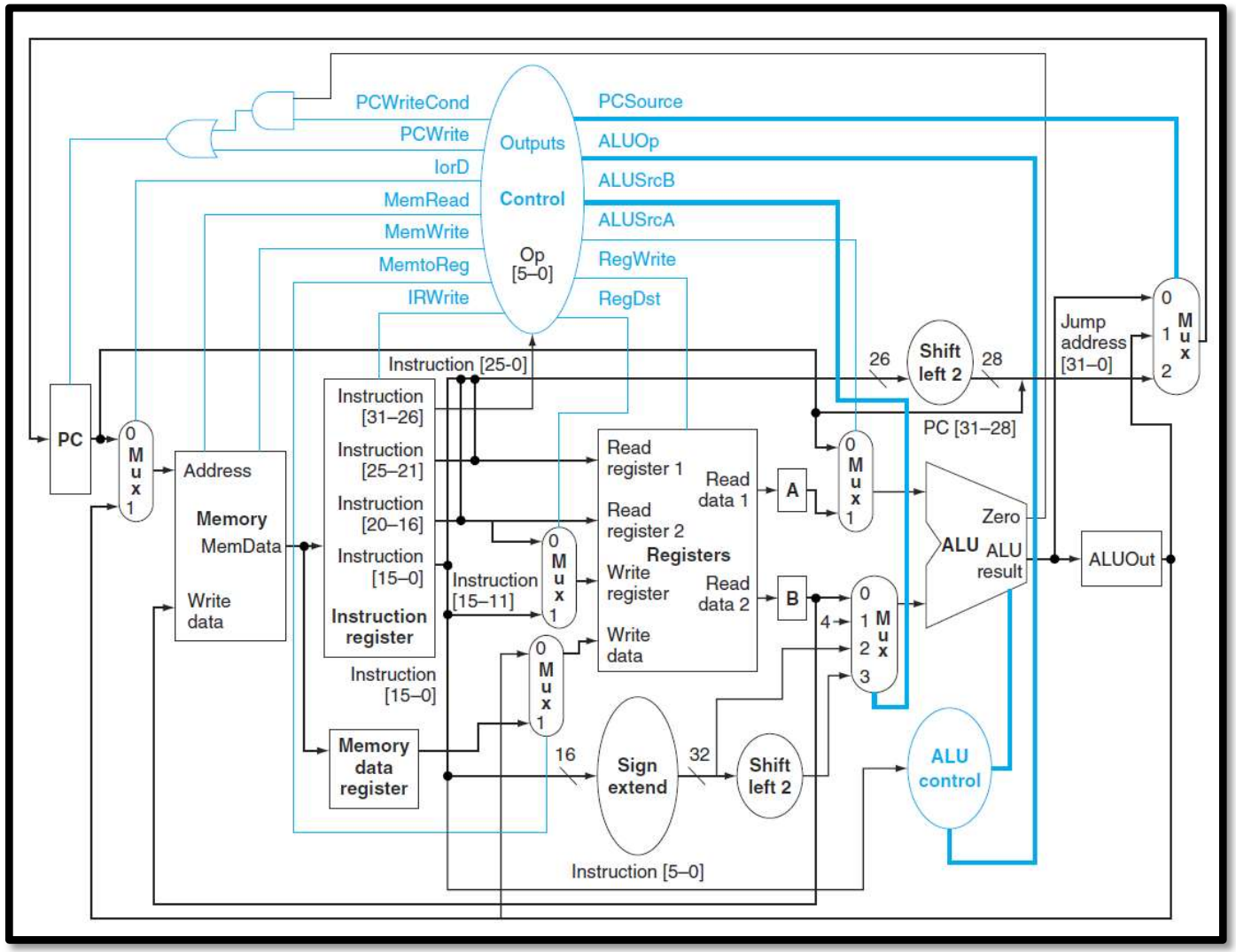
SRA, SLL

r_t << Shamt (Instr[10-6])

J-Type Instructions

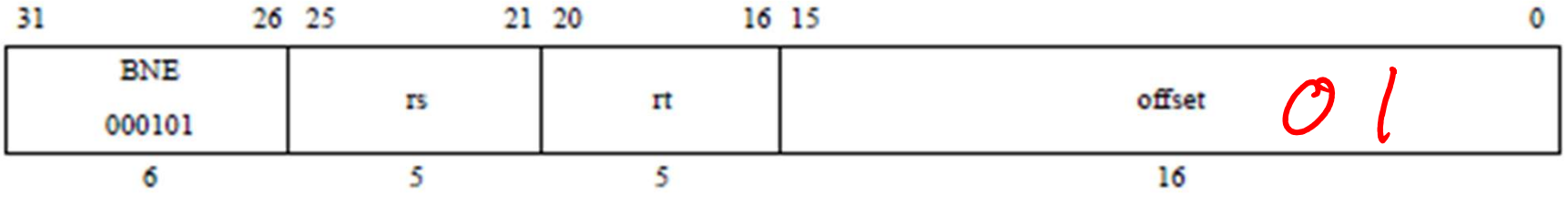
ADDI	ADDU	AND
ORI	SUB	SRA
SLLV	SLL	SLTI
LUI	SW	LW
LH	CLO	BNE
J	JR	MULTU
MFLO	MFHI	

Milestone 3 – Implement J-Type instructions



MIPS Multicycle CPU Datapath

BNE



Format: `BNE rs, rt, offset` MIPS32 (MIPS I)

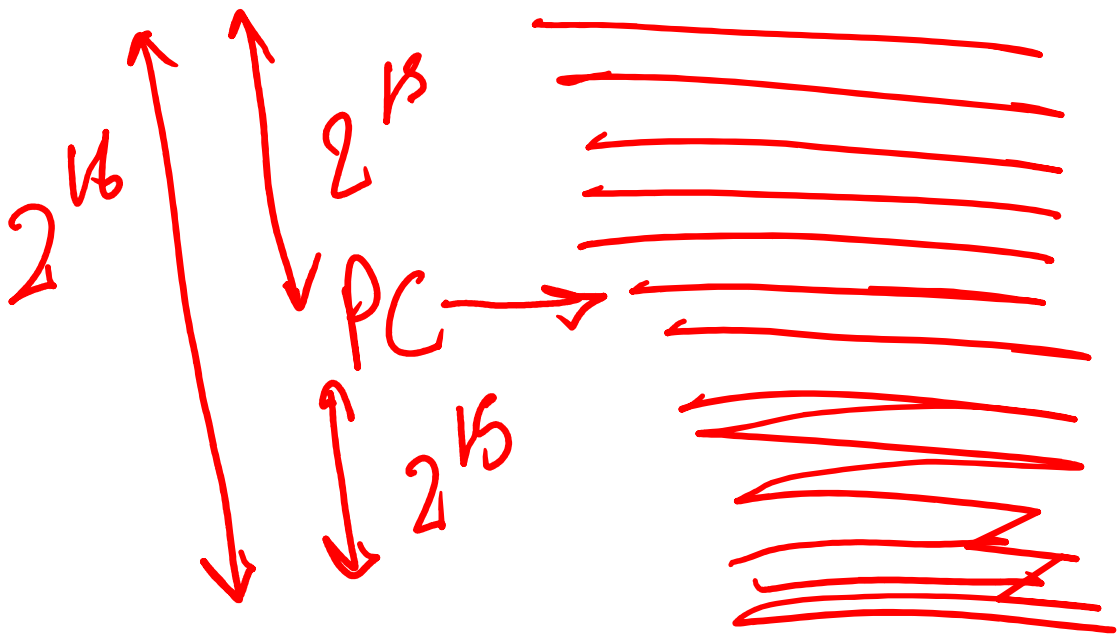
Purpose:

To compare GPRs then do a PC-relative conditional branch

Description: `if rs ≠ rt then branch`

An 18-bit signed offset (the 16-bit *offset* field shifted left 2 bits) is added to the address of the instruction following the branch (not the branch itself), in the branch delay slot, to form a PC-relative effective target address.

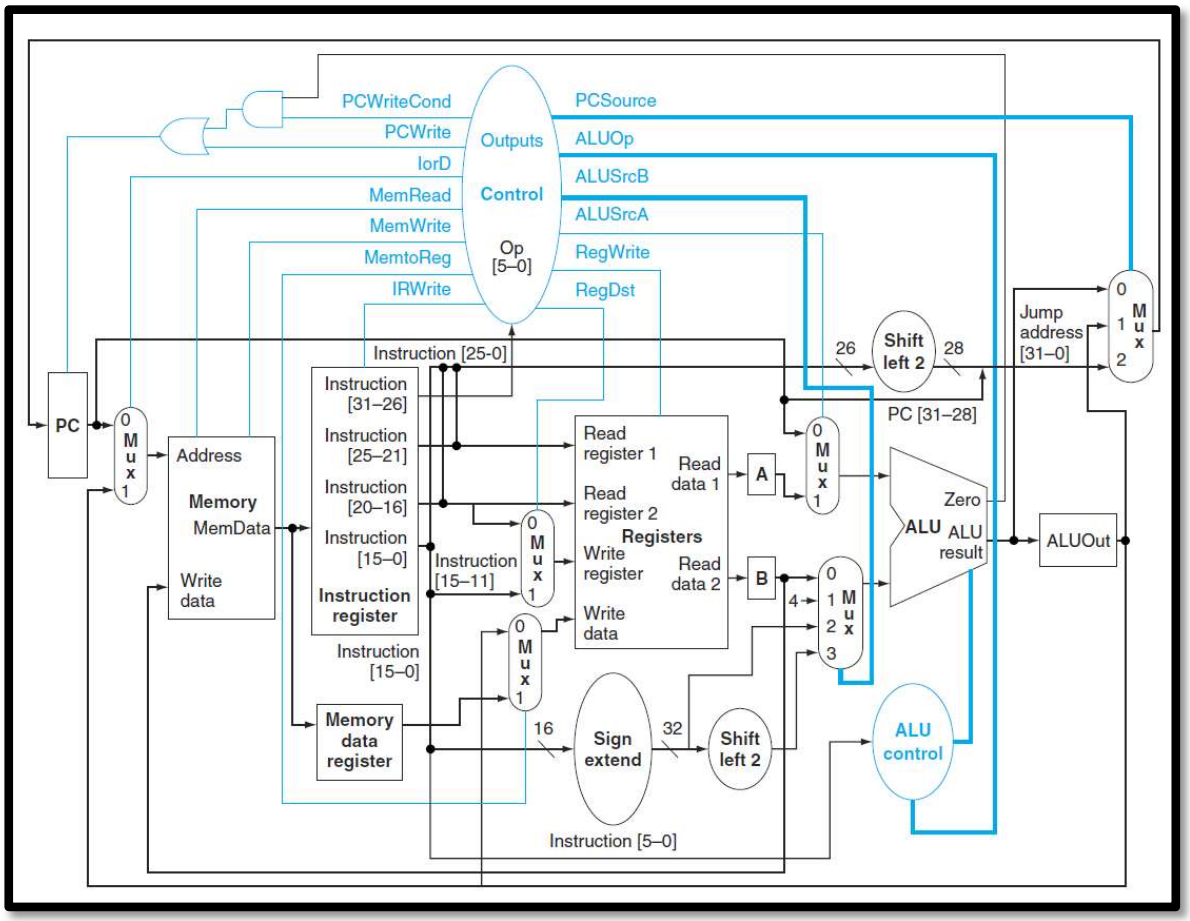
If the contents of GPR *rs* and GPR *rt* are not equal, branch to the effective target address after the instruction in the delay slot is executed.



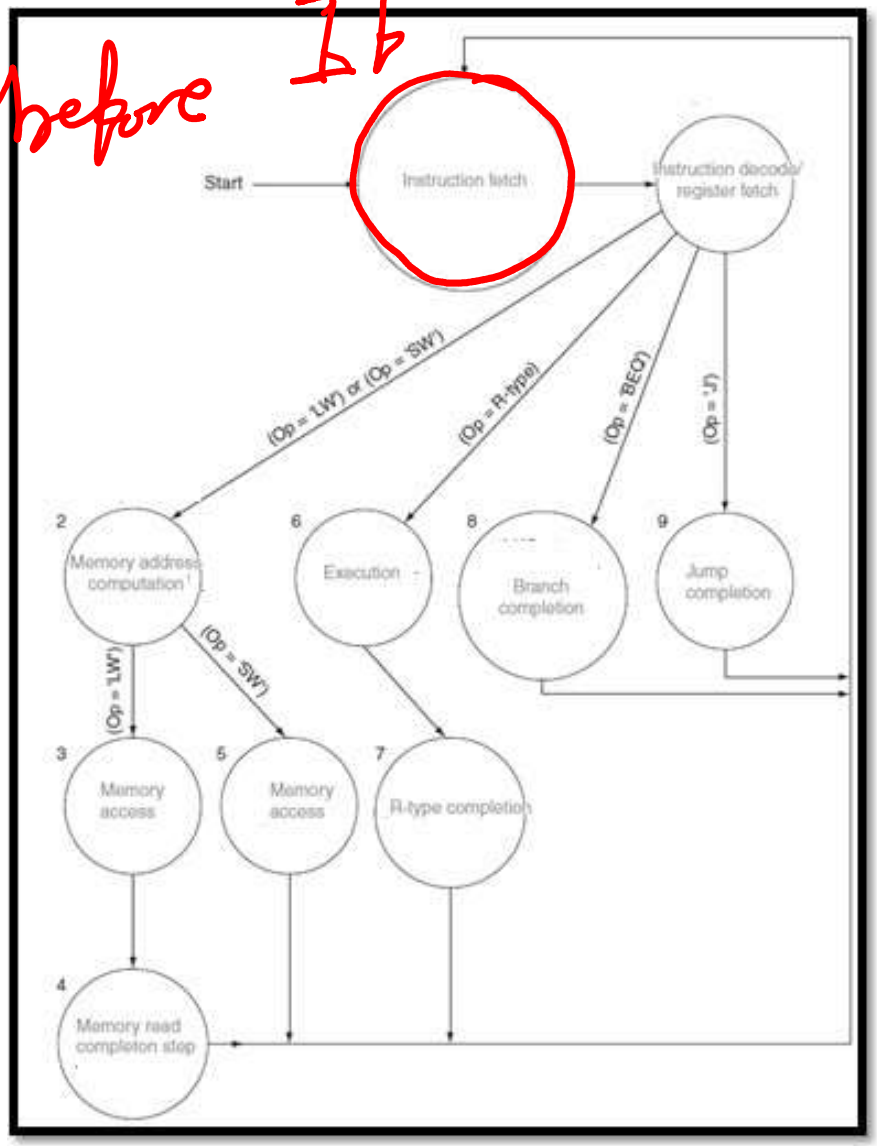
Datapath and Control

MIPS Multicycle CPU FSM

Same as before



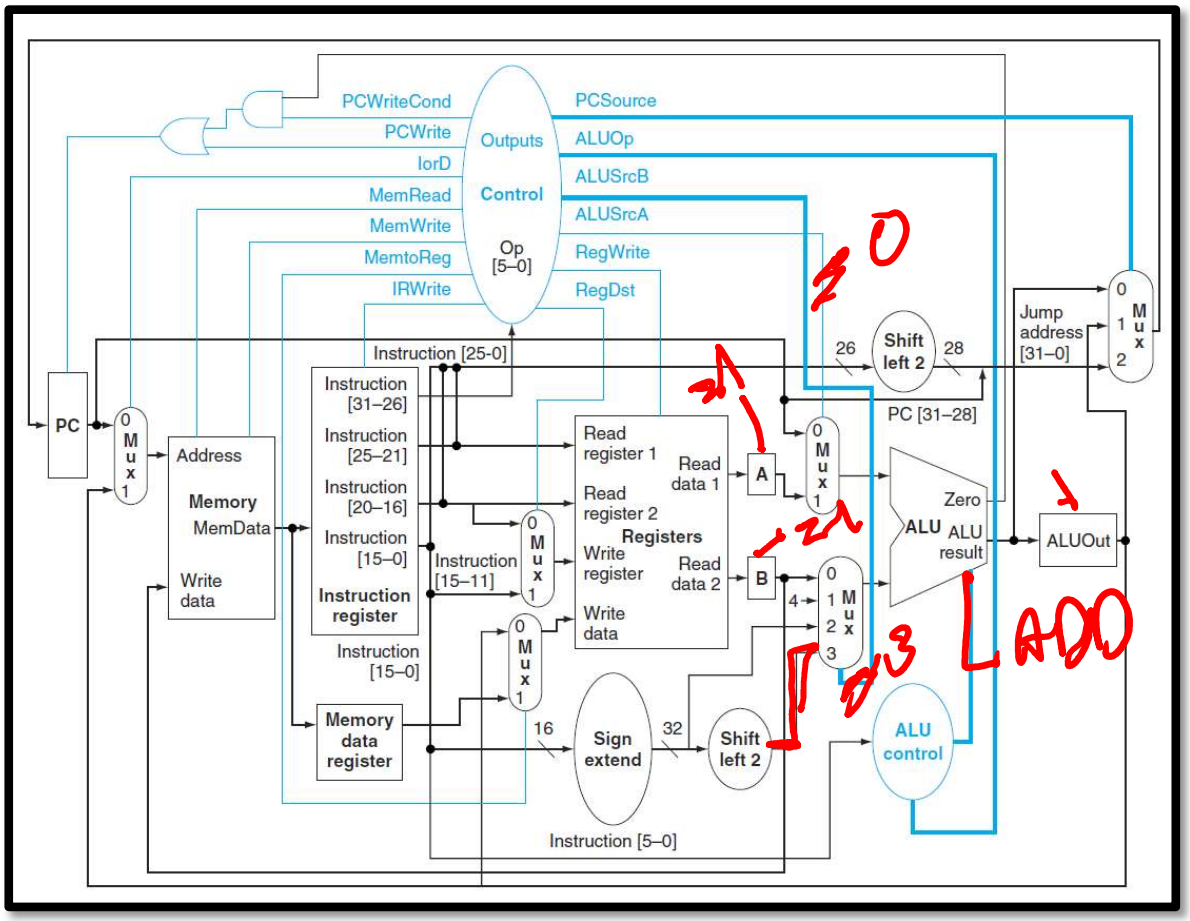
MIPS Multicycle CPU Datapath



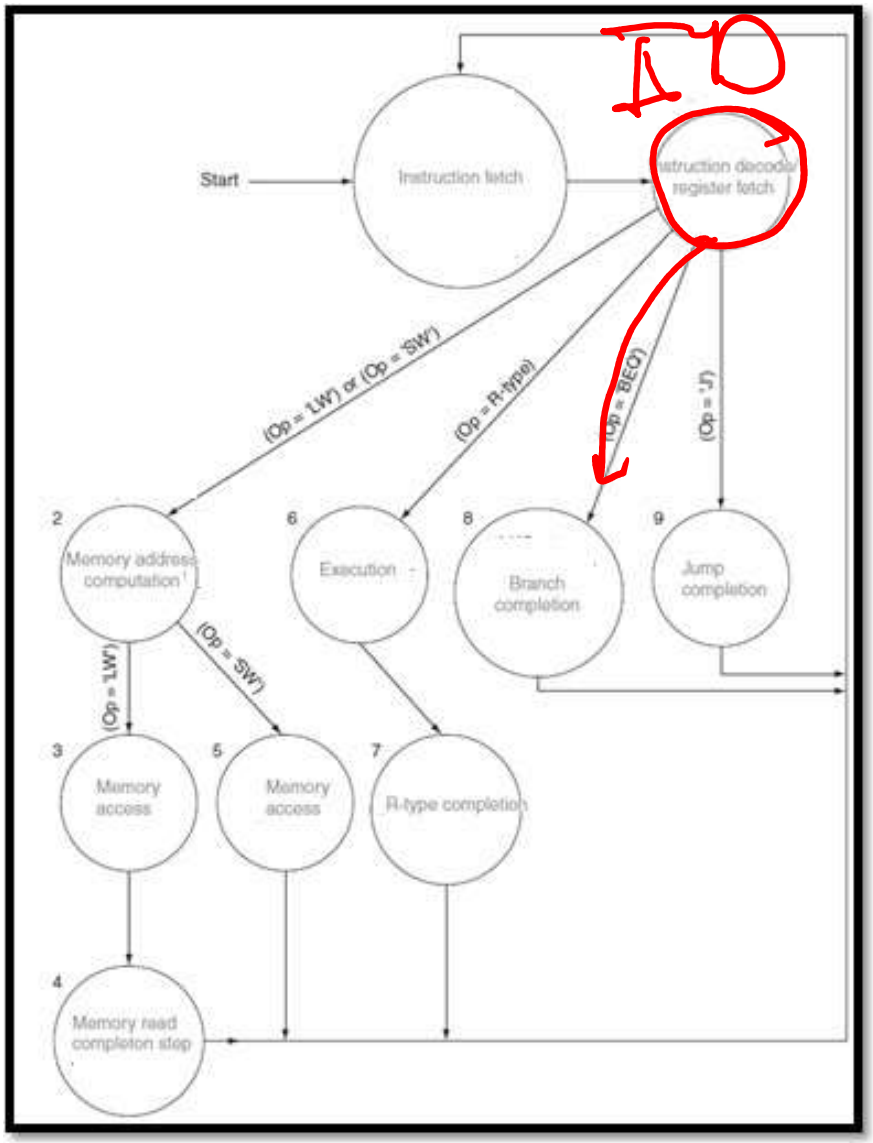
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



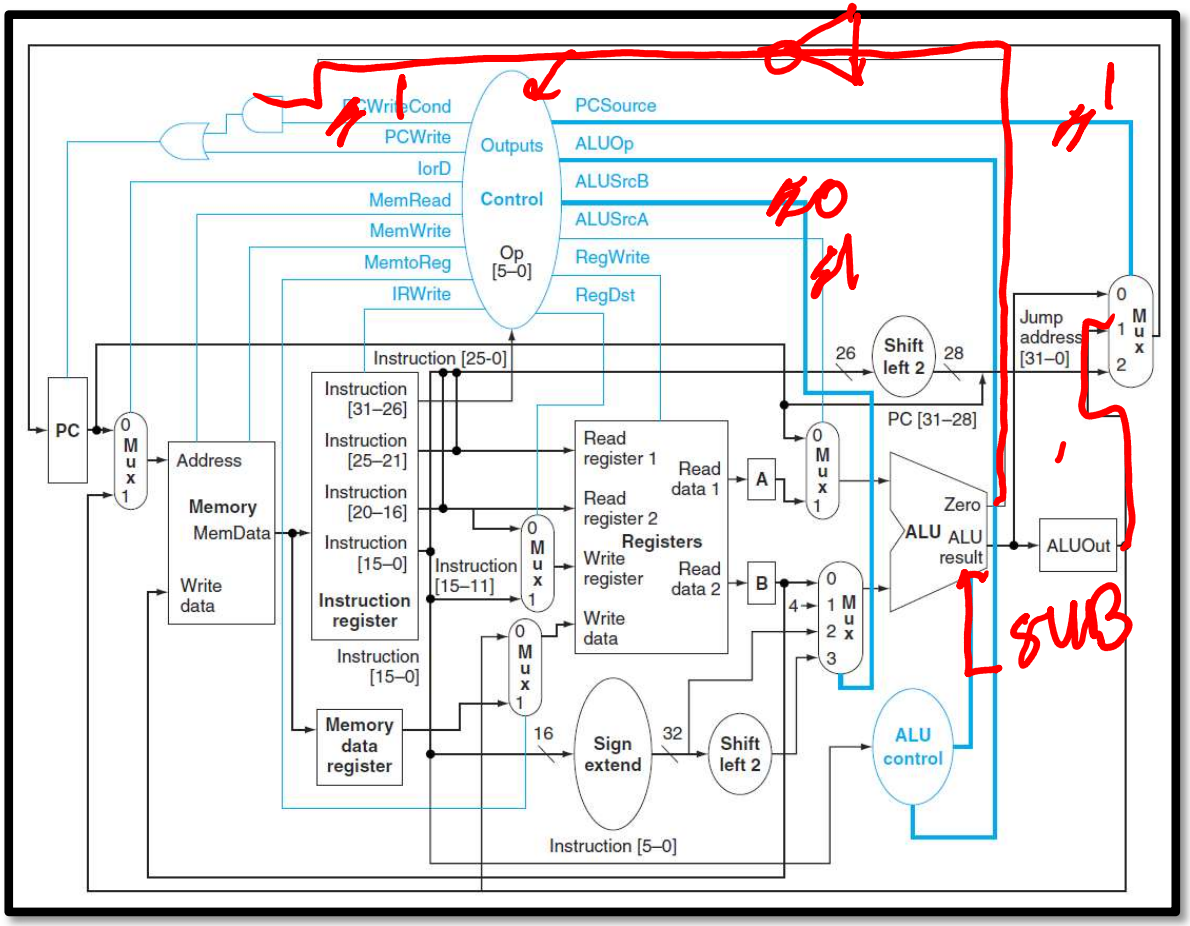
MIPS Multicycle CPU Datapath



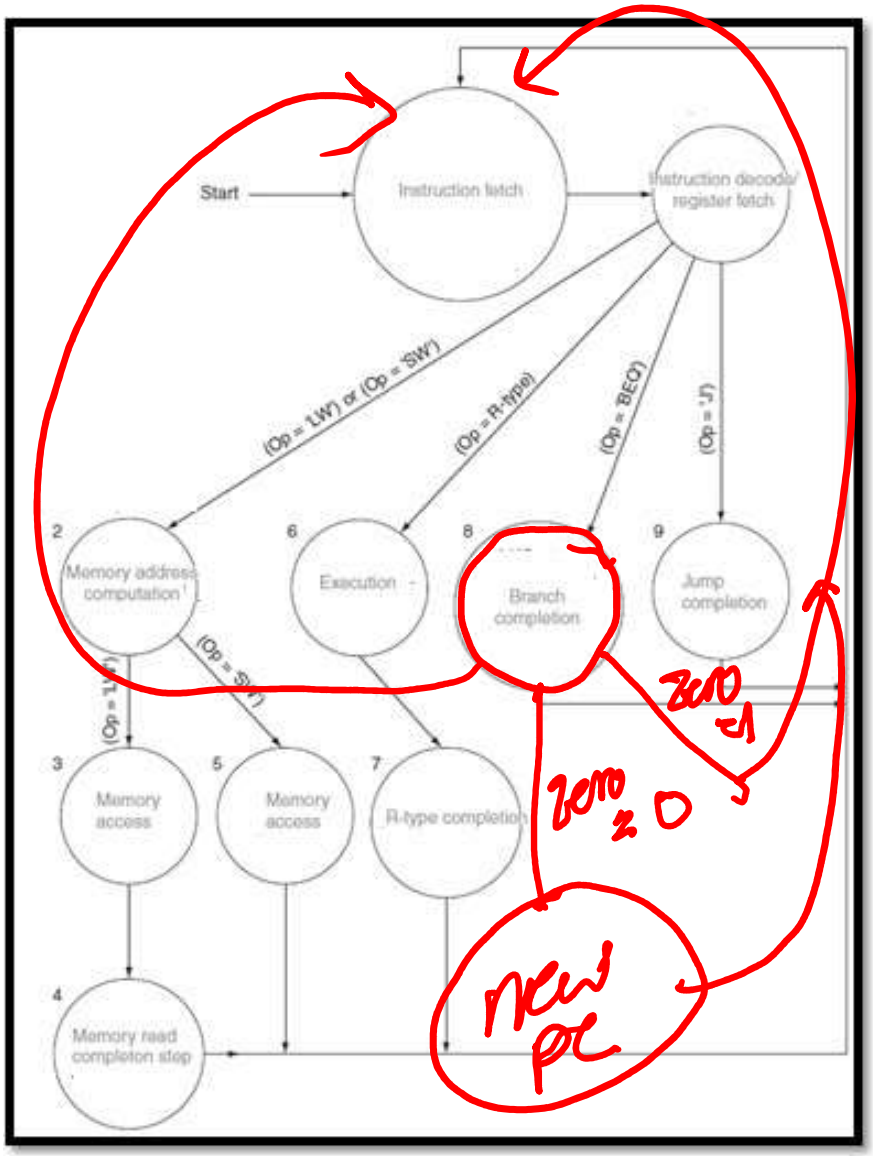
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



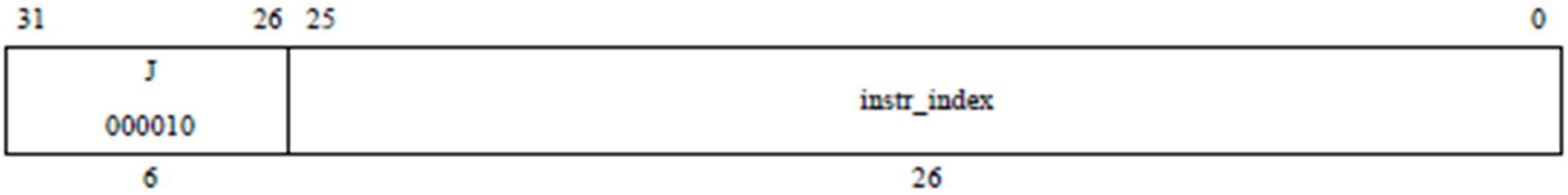
MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

J

Jump J



Format: `J target` MIPS32 (MIPS I)

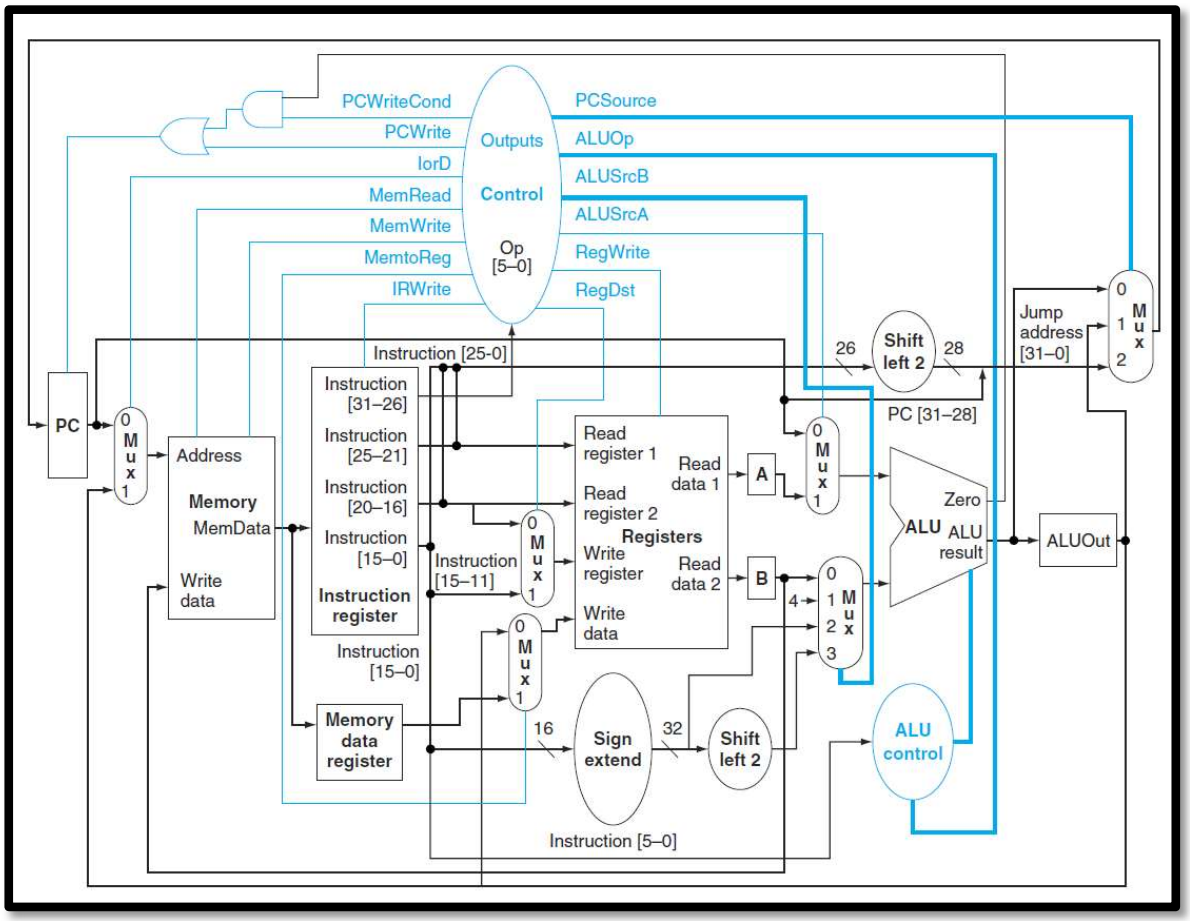
PC[31-28] & [(Instr[25-0]) << 2]



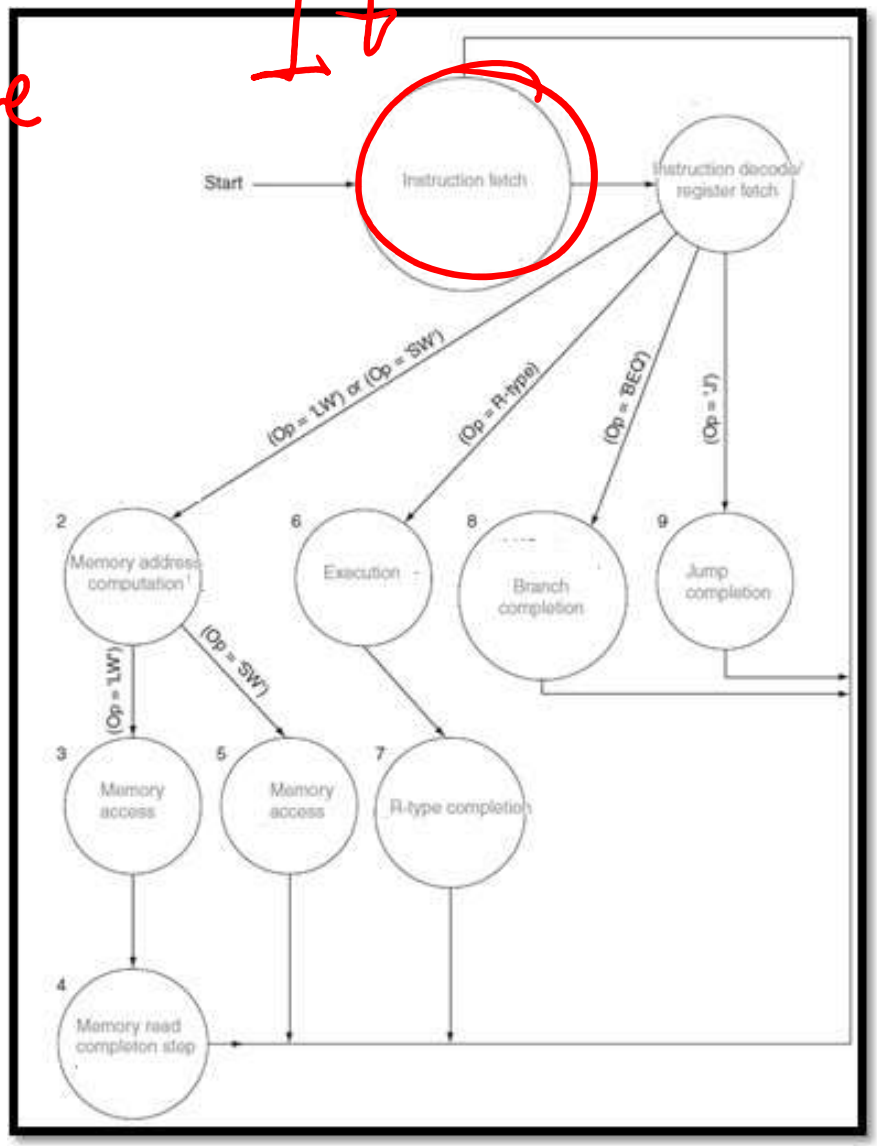
32 - bits

Datapath and Control

MIPS Multicycle CPU FSM



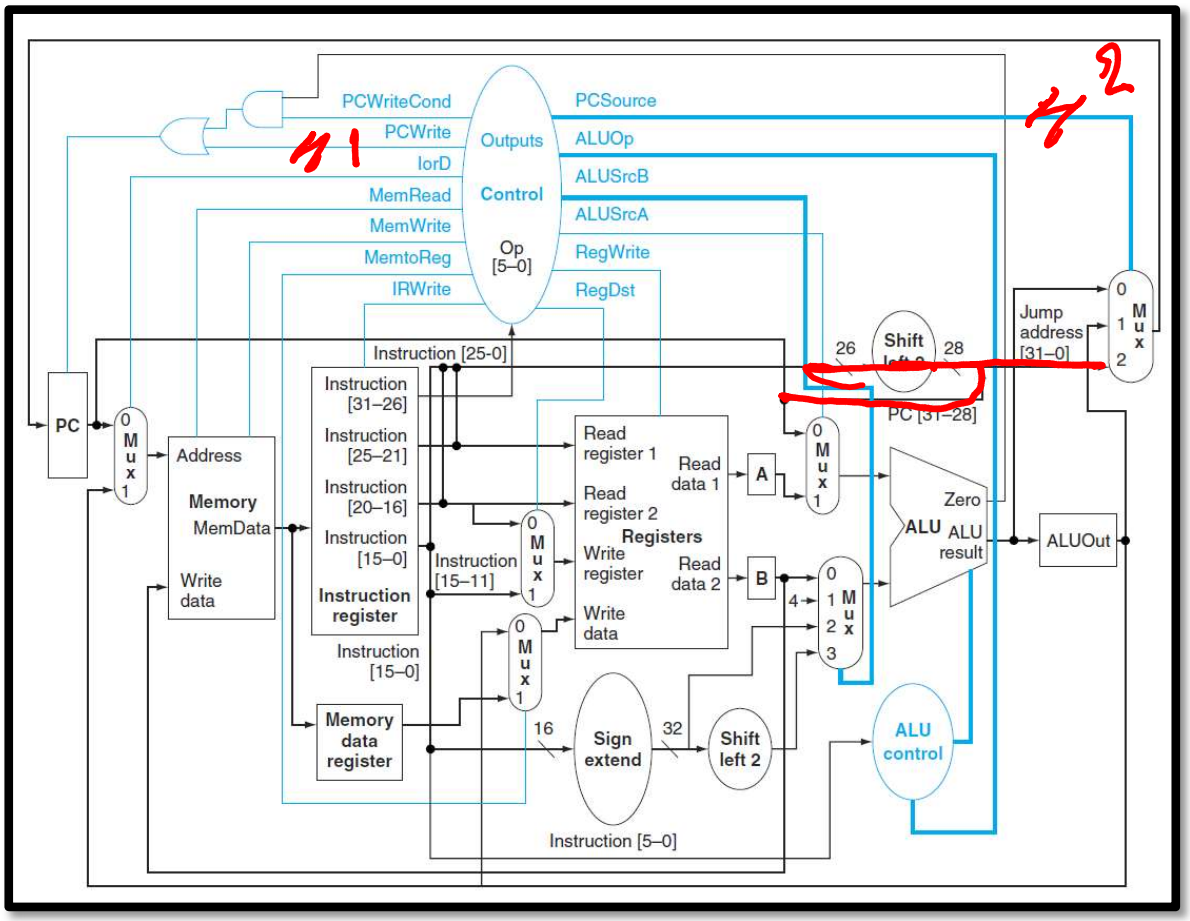
MIPS Multicycle CPU Datapath



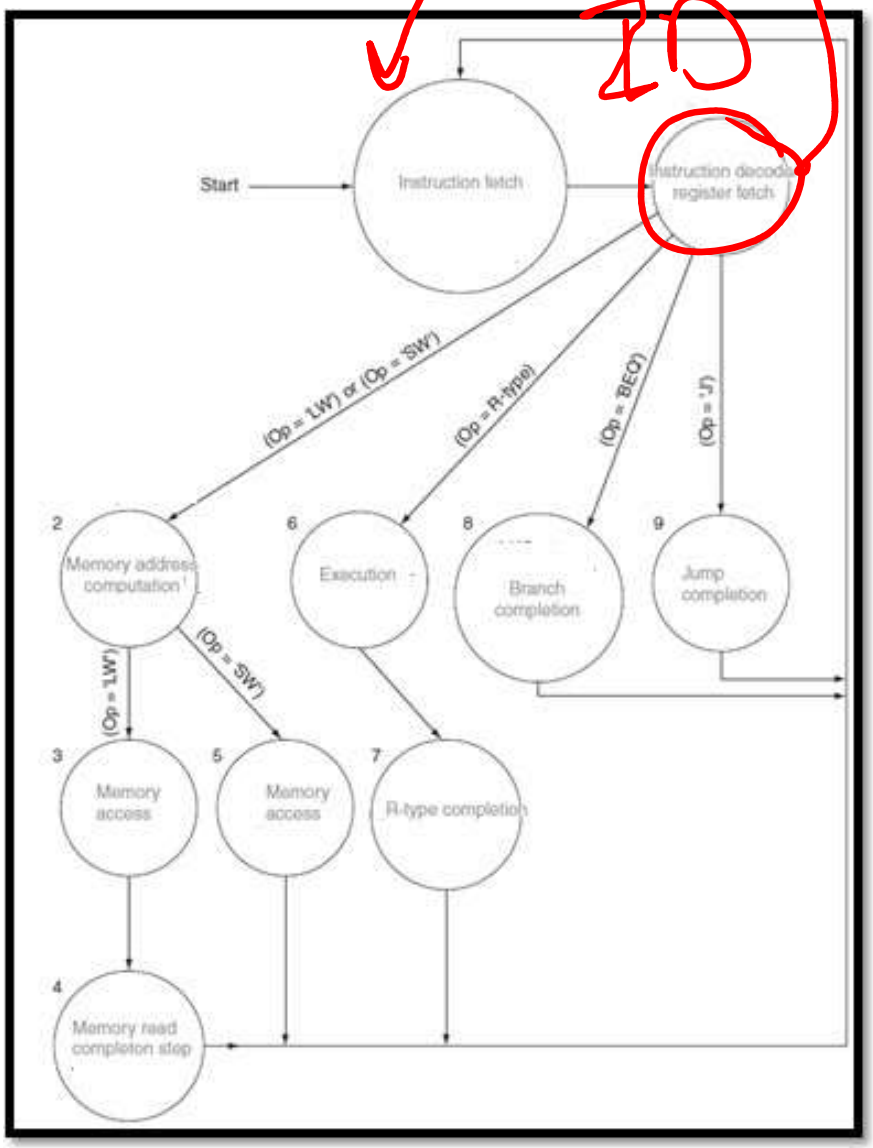
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM

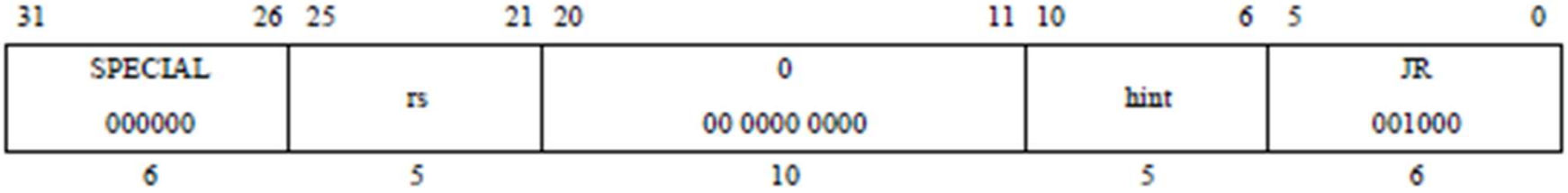


MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

JR



Format: `JR rs` MIPS32 (MIPS I)

Purpose:

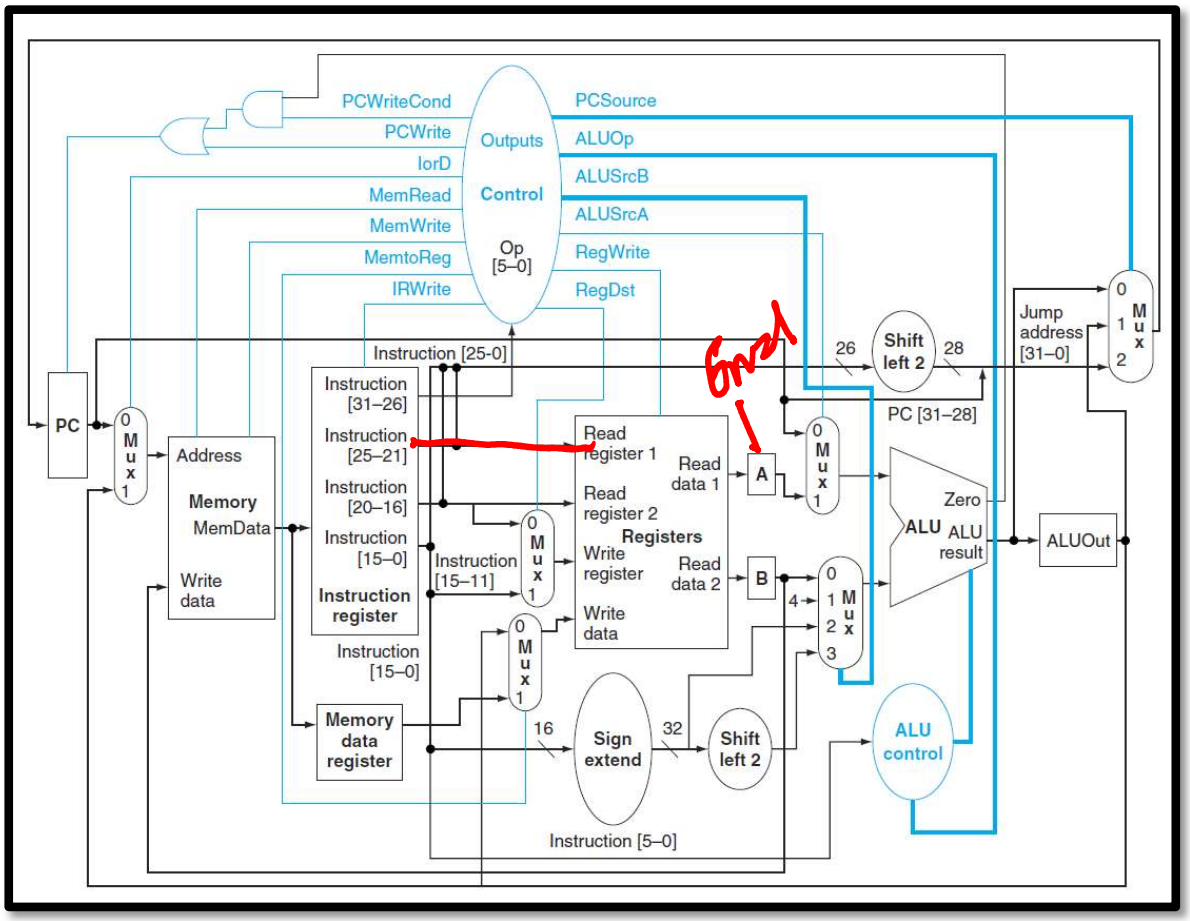
To execute a branch to an instruction address in a register

Description: `PC ← rs`

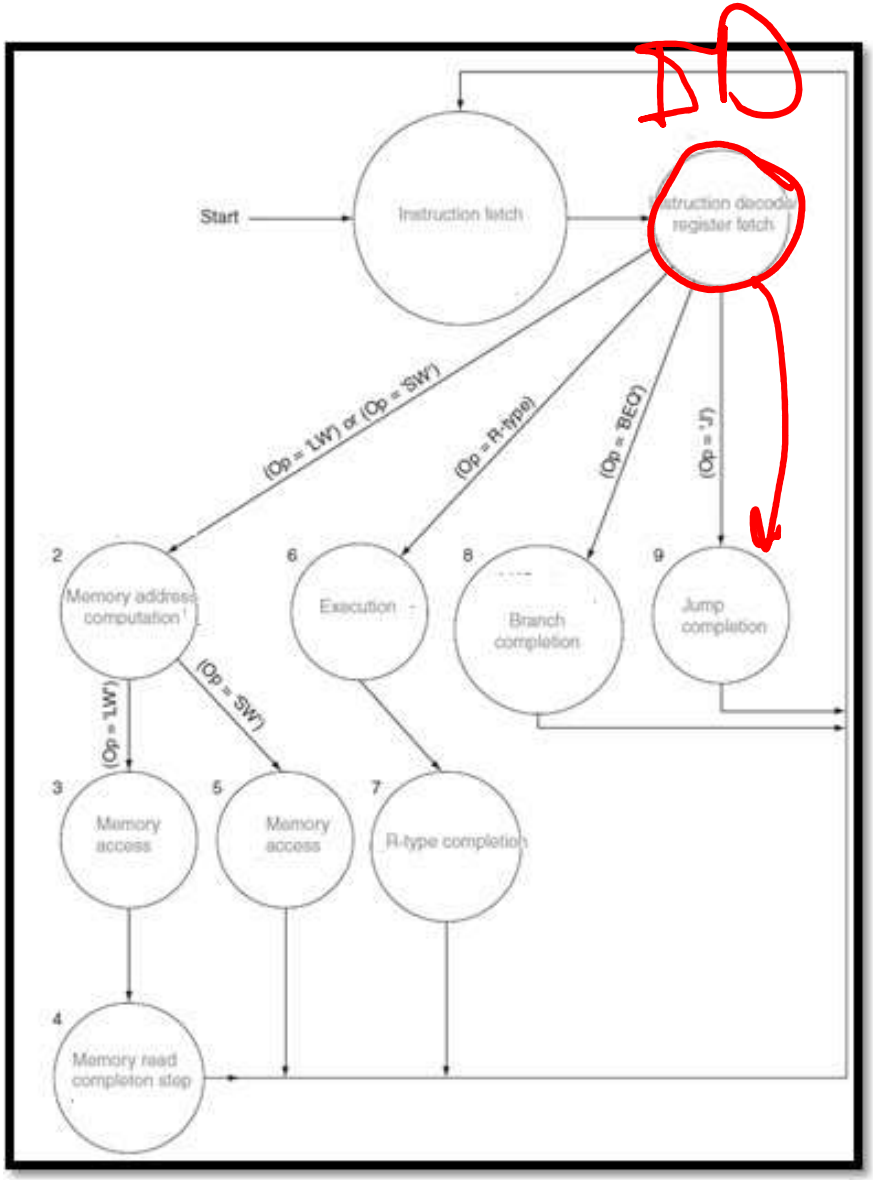
Jump to the effective target address in GPR *rs*. Execute the instruction following the jump, in the branch delay slot, before jumping.

Datapath and Control

MIPS Multicycle CPU FSM



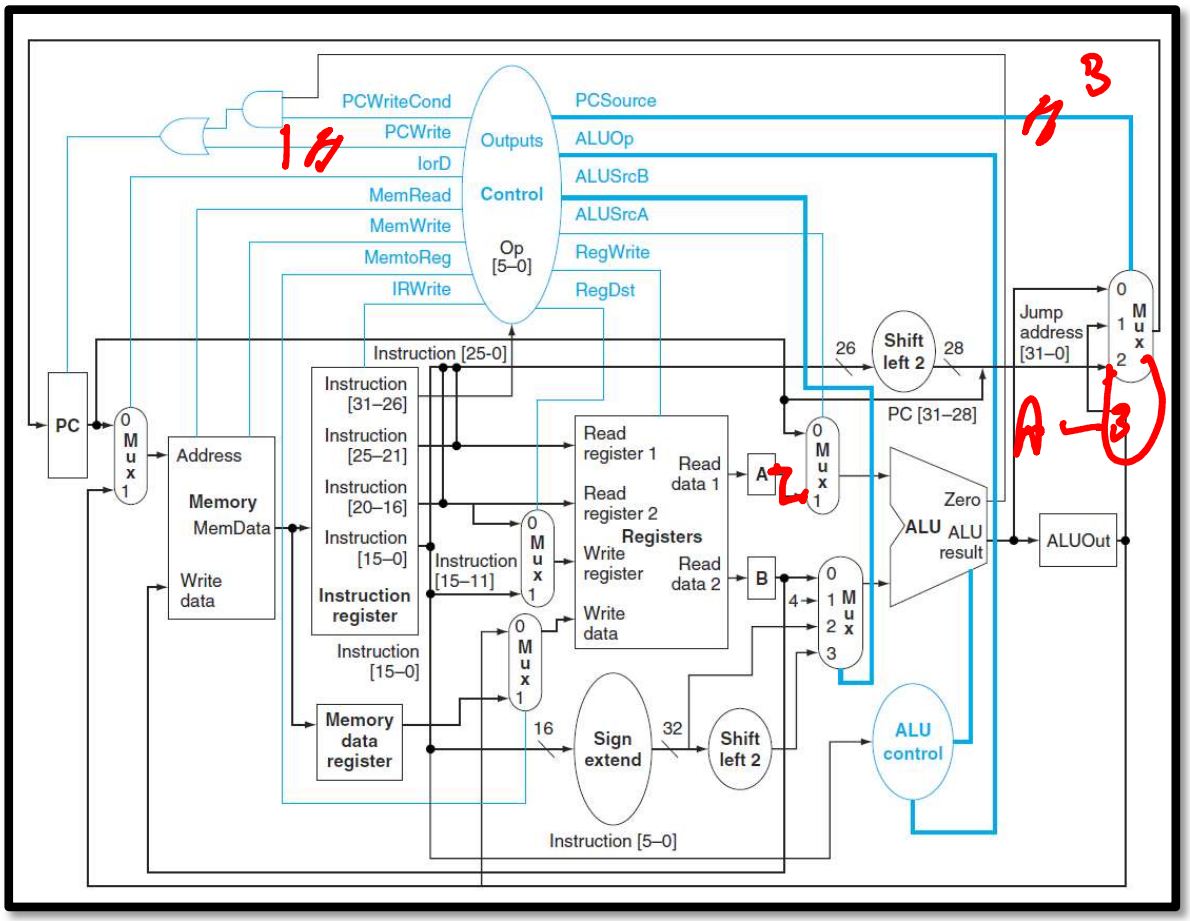
MIPS Multicycle CPU Datapath



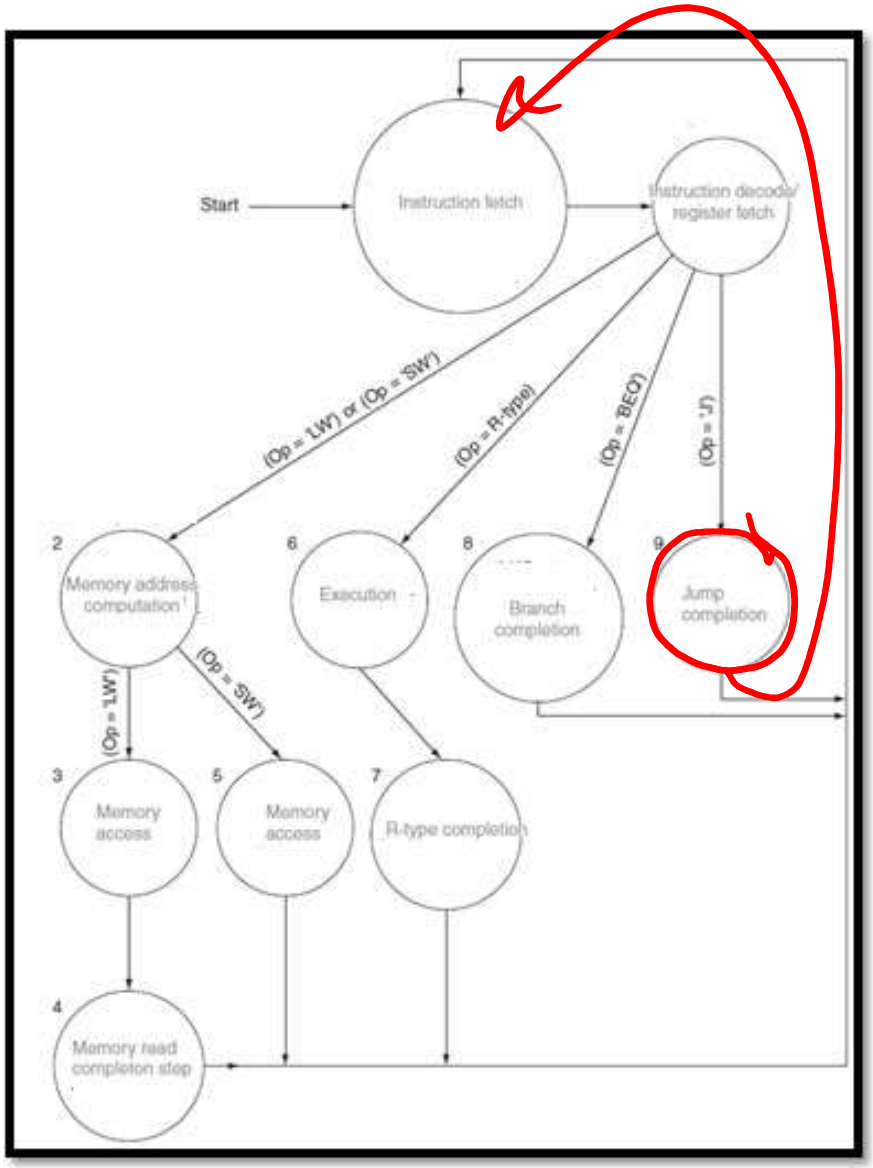
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller